

Problems

Problem 1

All the vertices of the triangle ABC lie on the graph $y = \frac{1}{x}$ on a cartesian plane. If its orthocenter lies on the point $(-1, -1)$ and its centroid lies on the point $(7, 9)$, the length of its circumradius can be expressed as $\sqrt{m}$. Find m .

Problem 2

A toy store sells building blocks with heights of 1, 2, 3, 4, and 5. Paul has an ample number of each type of block and wants to create two towers. How many possible ordered pairs of two towers with equal height can Paul create with exactly 9 building blocks so that the bases of the 7 blocks not touching the ground each have a unique elevation. (Note: Blocks of the same height are indistinguishable)

Problem 3

How many ways can an integer $1 - 14$ be placed at every vertex of a 13-gon (they don't necessarily need to be distinct) so that no substring of consecutive vertices have a total sum divisible by 15

Problem 4

Let ABC be an acute triangle so that the distances from its circumcenter, orthocenter, and incenter to segment BC are 6, 3, and 5 respectively. If the area of ABC can be expressed as $m\sqrt{n}$ for positive integers m and n where n is square-free, find $m + n$

Problem 5

Let ABC be a triangle with circumcenter O and let D be a point on BC . Suppose line BO meets the circumcenter of ABD at E and let line CO meet the circumcenter of ACD at F . If the circumcenter of DEF is X , prove that $XO \parallel BC$

Problem 6

Let $ABCD$ be a rectangle and let ω be the circle with center A that passes through C . If line BD intersects ω at points P and Q so that $PB < PD$. Given that $PB = 10$ and $DQ = 12$, the area of $ABCD$ can be written as $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Problem 7

Let N be the number of 20-digit numbers that exist so that exactly 8 pairs of adjacent digits sum to a multiple of 3, exactly 8 pairs of adjacent digits differ by a multiple of 3, and exactly 7 pairs of adjacent digits multiply to a multiple of 3. Find the number of divisors of N

Problem 8

Let ABC be an acute triangle and let X and Y be points on AB and AC respectively so that the circumcircle of triangle AXY is tangent to BC at the foot of the A -altitude. Suppose the B -altitude and C -altitude intersect XY at points P and Q so that $XP = 1$, $PQ = 2$, and $QY = 6$. Find the

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remainder when the square of the area of ABC is divided by 1000.

Problem 9

For a real number x , let $\lfloor x \rfloor$ be the greatest integer less than or equal to x , and define $x - \lfloor x \rfloor$ to be the fractional part of x . For a positive integer n , the infinite sequence of numbers

$$\frac{n}{2}, \frac{n}{2^2}, \frac{n}{2^3}, \frac{n}{2^4}, \dots$$

is written on a whiteboard. Upon noticing the whiteboard, Bob sums the fractional parts of every term of the sequence and Joe sums the fractional part of every third term in the sequence (starting with $\frac{n}{2^3}$). Given that Bob's sum is twice Joe's sum, find the smallest possible value of n .

Problem 10

Let $AIME$ be a rectangle and let I, N, D, E, X be colinear points in that order such that $IN = 3$, $ND = 1$, $DE = 5$. Given that (MDX) bisects AE and (ANX) bisects ME , the area of $AIME$ can be written in simplest form as $m\sqrt{n}$. Find $m + n$

Problem 11

Three points A , B , and C lie on the graph of $y = x^2$. Given that lines AB , BC , CA contain the points $(1, -7)$, $(1, 4)$, and $(1, 7)$ respectively, the area of ABC can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$

Problem 12

Let X be the number of ordered pairs of monic polynomials with integer coefficients $(P(x), Q(x))$ such that

$$P(x) \cdot Q(x) = (x - 1)(x - 2)(x - 3) \dots (x - 17)$$

and the equation $\max(P(x), Q(x)) = 0$ has exactly 10 real solutions. Compute the remainder when X is divided by 1000.

Problem 13

There are 21 students standing in a line. Every hour, a random pair of adjacent students have a fight, causing the losing student to exit the line. Given that each student has an equal chance of winning each fight, the probability that the final fight is between the third student from the left and the fourth student from the right in the initial line can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.

Problem 14

Define a sequence of rational numbers, $a_0, a_1, a_2, \dots$ so that $a_{k+1} = |1 - \frac{1}{a_k}|$. Find the number of possible values of a_0 so that $a_{12} = 0$.

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Problem 15

Find the product of all possible real values of k so that there exists a non-zero polynomial $P(x)$ such that

$$P(x) = P(20)x^2 + P(k)x + P(23)$$

Problem 16

Let S be the set of the first 8 prime numbers. For each subset $\mathcal{T}$ of S , let $f(\mathcal{T})$ be the remainder when the product of the elements of $\mathcal{T}$ is divided by 6. Find the sum of $f(\mathcal{T})$ over all subsets of S . Note: for the empty set, we define $f(\emptyset) = 1$.

Problem 17

Let ABC be a triangle with circumcenter O . Suppose D is the foot of the altitude from A to BC . If the distances from O to AB , AD , and AC are 7, 5, and 3 respectively, find the area of ABC .

Problem 18

How many ways to color each edge of 4×4 grid with one of 6 colors such that each cell has exactly 3 colors on its perimeter

Problem 19

Distinct real numbers, r_1 , r_2 and r_3 are roots of the equation $12r_1x^3 + 72r_2x^2 + 432r_3x = 0$. Find the value of $r_1^2 + r_2^2 + r_3^2$.

Problem 20

There exist quadratic functions $P(x)$ and $Q(x)$ with real coefficients. Each of the three equations

$$P(x) = Q(x)$$

$$P(x) = 2Q(x)$$

$$P(x) = 3Q(x)$$

has exactly 1 solution. Given that the solutions to these three equations are $x = 20$, 22 , and n for a real number n , find the sum of all possible values of n

Problem 21

Let $ABCD$ be a rhombus. Suppose the ellipse with foci at A and C that pass through B and D has area $\sqrt{71}\pi$ and the ellipse with foci B and D that pass through A and C has area $\sqrt{125}\pi$. Find the square of the perimeter of $ABCD$

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Problem 22

Let $P(x)$ be a quadratic polynomial such that the product of the roots of P is 20. Real numbers a and b satisfy $a + b = 22$ and $P(a) + P(b) = P(22)$. Find $a^2 + b^2$.

Problem 23

There are 10 distinguishable penguins numbered from 1 to 10 sitting in a row of 10 seats. One day, Fred adds 3 empty seats to the left of the row and then goes to sleep. Every subsequent day, one penguin moves from its current seat to an empty seat to the left. After many days, Fred returns, and finds that the penguins now occupy the 10 leftmost seats in the row. Let N be the number of possible seating arrangements the penguins could be in. Compute the remainder when N is divided by 1000.

Problem 24

Let DEF be the orthic triangle of ABC . Given that $[BDF] = 10$, $[DEF] = 15$, $[CDE] = 40$, find $\tan \angle BAC$

Problem 25

There exist five basketball teams, each with a different fixed skill level such that when two of the teams play a basketball game, the team with the higher skill level always wins. Every day a manager randomly chooses two teams to play a game with each other. Let x be the expected number of games the manager needs to observe in order to correctly rank the teams' skill levels. If x can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , what is $m + n$?

Problem 26

Let ABC be a triangle and let D , E , and F be points on segments BC , AC , and AB respectively, so that E is the reflection of B over DF , and F is the reflection of C over DE . Segments CF and BE meet at a point G . Given that $AG = 8$ and $EF = 7$, find the area of triangle EGF

Problem 28

An angle θ is randomly chosen from the interval $[0, 2\pi]$. The probability that $\sin(3\theta) > \sin(2\theta) > \sin(\theta)$ can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find $m + n$.

Problem 29

Every minute, Juan chooses a random divisor of 2310 and writes it on a blackboard. He continues process indefinitely until two numbers on the board are not relatively prime. Find the number of ways that Juan can do this so that he stops when exactly 4 numbers are written on the board.